

KUIS 3

Mata Kuliah : Kalkulus 2
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Sifat : Open Cheat Sheet
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Diberikan 5 soal dengan bobot nilai yang sama dan boleh dikerjakan tidak berurutan.
Tuliskan: Nama, NRP, dan Nomor Kelas pada lembar jawaban Anda.

DILARANG MELAKUKAN TINDAK KECURANGAN WAKTU UJIAN
“Setiap tindak kecurangan akan mendapatkan sanksi akademik.”

1. Hitung integral berikut dengan integral parsial:

$$\int e^{7x} x^2 dx.$$

2. Dengan menggunakan rumus reduksi, hitung integral berikut:

(a)

$$\int \cos^6(2x) \sin^5(2x) dx.$$

(b)

$$\int \tan^6 \theta \sec^3 \theta d\theta.$$

3. Dengan integrasi pecahan parsial, hitunglah integral

$$\int \frac{3x^2 - x}{(x - 2)(x^2 + 1)} dx$$

4. Dengan menggunakan substitusi Trigonometri, hitunglah Integral dari

$$\int x^2 \sqrt{x^2 - 16} dx$$

5. Dengan substitusi bentuk pangkat rasional, hitunglah

$$\int \frac{t^{3/2}}{t + t^{4/3}} dt$$

SOLUSI

1. Integral Parsial

Misal $u = x^2 \implies du = 2x dx$

dan $dv = e^{7x} dx \implies v = \frac{1}{7}e^{7x}$.

$$\begin{aligned}\int e^{7x} x^2 dx &= uv - \int v du \\ &= \frac{1}{7} x^2 e^{7x} - \frac{2}{7} \int x e^{7x} dx\end{aligned}$$

Gunakan integral parsial lagi untuk $\int x e^{7x} dx$:

Misal $u = x \implies du = dx$

dan $dv = e^{7x} dx \implies v = \frac{1}{7}e^{7x}$.

$$\begin{aligned}\int x e^{7x} dx &= \frac{1}{7} x e^{7x} - \int \frac{1}{7} e^{7x} dx \\ &= \frac{1}{7} x e^{7x} - \frac{1}{49} e^{7x}\end{aligned}$$

Substitusi kembali ke persamaan awal:

$$\begin{aligned}\int e^{7x} x^2 dx &= \frac{1}{7} x^2 e^{7x} - \frac{2}{7} \left(\frac{1}{7} x e^{7x} - \frac{1}{49} e^{7x} \right) + C \\ &= \frac{1}{7} x^2 e^{7x} - \frac{2}{49} x e^{7x} + \frac{2}{343} e^{7x} + C\end{aligned}$$

2. Integral Trigonometri & Rumus Reduksi

(a) $\int \cos^6(2x) \sin^5(2x) dx$

Pisahkan satu $\sin(2x)$:

$$\int \cos^6(2x) (\sin^2(2x))^2 \sin(2x) dx = \int \cos^6(2x) (1 - \cos^2(2x))^2 \sin(2x) dx$$

Misal $u = \cos(2x) \implies du = -2 \sin(2x) dx \implies \sin(2x) dx = -\frac{1}{2} du$.

$$\begin{aligned}\int u^6 (1 - u^2)^2 \left(-\frac{1}{2} \right) du &= -\frac{1}{2} \int u^6 (1 - 2u^2 + u^4) du \\ &= -\frac{1}{2} \int (u^6 - 2u^8 + u^{10}) du \\ &= -\frac{1}{2} \left[\frac{u^7}{7} - \frac{2u^9}{9} + \frac{u^{11}}{11} \right] + C \\ &= -\frac{1}{14} \cos^7(2x) + \frac{1}{9} \cos^9(2x) - \frac{1}{22} \cos^{11}(2x) + C\end{aligned}$$

(b) $\int \tan^6 \theta \sec^3 \theta d\theta$

Gunakan identitas $\tan^2 \theta = \sec^2 \theta - 1$:

$$\int (\sec^2 \theta - 1)^3 \sec^3 \theta d\theta = \int (\sec^6 \theta - 3\sec^4 \theta + 3\sec^2 \theta - 1) \sec^3 \theta d\theta$$

$$= \int (\sec^9 \theta - 3 \sec^7 \theta + 3 \sec^5 \theta - \sec^3 \theta) d\theta$$

Gunakan rumus reduksi $\int \sec^n \theta d\theta = \frac{\sec^{n-2} \theta \tan \theta}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} \theta d\theta$ secara berulang untuk menyelesaikan integral dari tiap suku tersebut.

3. Pecahan Parsial

$$\frac{3x^2 - x}{(x-2)(x^2+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1}$$

Kalikan dengan penyebut:

$$3x^2 - x = A(x^2 + 1) + (Bx + C)(x - 2)$$

Substitusi $x = 2 \implies 12 - 2 = A(5) \implies 10 = 5A \implies A = 2$.

Substitusi $x = 0 \implies 0 = 2(1) + C(-2) \implies 2C = 2 \implies C = 1$.

Substitusi $x = 1 \implies 2 = 2(2) + (B+1)(-1) \implies 2 = 4 - B - 1 \implies B = 1$.

Sehingga integralnya menjadi:

$$\begin{aligned} \int \left(\frac{2}{x-2} + \frac{x+1}{x^2+1} \right) dx &= \int \frac{2}{x-2} dx + \int \frac{x}{x^2+1} dx + \int \frac{1}{x^2+1} dx \\ &= 2 \ln |x-2| + \frac{1}{2} \ln(x^2+1) + \arctan(x) + C \end{aligned}$$

4. Substitusi Trigonometri

Bentuk $\sqrt{x^2 - 16}$, gunakan $x = 4 \sec \theta \implies dx = 4 \sec \theta \tan \theta d\theta$, dan $\sqrt{x^2 - 16} = 4 \tan \theta$.

$$\begin{aligned} \int x^2 \sqrt{x^2 - 16} dx &= \int (16 \sec^2 \theta)(4 \tan \theta)(4 \sec \theta \tan \theta) d\theta \\ &= 256 \int \sec^3 \theta \tan^2 \theta d\theta \\ &= 256 \int \sec^3 \theta (\sec^2 \theta - 1) d\theta \\ &= 256 \int (\sec^5 \theta - \sec^3 \theta) d\theta \\ &= 256 \left[\left(\frac{1}{4} \sec^3 \theta \tan \theta + \frac{3}{4} \int \sec^3 \theta d\theta \right) - \int \sec^3 \theta d\theta \right] \\ &= 256 \left[\frac{1}{4} \sec^3 \theta \tan \theta - \frac{1}{4} \int \sec^3 \theta d\theta \right] \\ &= 64 \sec^3 \theta \tan \theta - 64 \left(\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right) \\ &= 64 \sec^3 \theta \tan \theta - 32 \sec \theta \tan \theta - 32 \ln |\sec \theta + \tan \theta| + C \end{aligned}$$

Kembalikan ke variabel x : $\sec \theta = \frac{x}{4}$ dan $\tan \theta = \frac{\sqrt{x^2-16}}{4}$.

$$\begin{aligned} &= 64 \left(\frac{x}{4} \right)^3 \left(\frac{\sqrt{x^2-16}}{4} \right) - 32 \left(\frac{x}{4} \right) \left(\frac{\sqrt{x^2-16}}{4} \right) - 32 \ln \left| \frac{x}{4} + \frac{\sqrt{x^2-16}}{4} \right| + C \\ &= \frac{1}{4} x^3 \sqrt{x^2-16} - 2x \sqrt{x^2-16} - 32 \ln |x + \sqrt{x^2-16}| + C' \end{aligned}$$

5. Substitusi Bentuk Pangkat Rasional

Penyebut pangkat pecahan memiliki penyebut 2 dan 3. KPK-nya adalah 6.

Misal $t = u^6 \implies dt = 6u^5 du$.

$t^{3/2} = (u^6)^{3/2} = u^9$, dan $t^{4/3} = (u^6)^{4/3} = u^8$.

$$\begin{aligned}\int \frac{t^{3/2}}{t + t^{4/3}} dt &= \int \frac{u^9}{u^6 + u^8} (6u^5) du \\ &= \int \frac{6u^{14}}{u^6(1 + u^2)} du \\ &= \int \frac{6u^8}{u^2 + 1} du\end{aligned}$$

Gunakan pembagian polinomial $\frac{u^8}{u^2+1} = u^6 - u^4 + u^2 - 1 + \frac{1}{u^2+1}$:

$$\begin{aligned}&= 6 \int \left(u^6 - u^4 + u^2 - 1 + \frac{1}{u^2 + 1} \right) du \\ &= 6 \left[\frac{u^7}{7} - \frac{u^5}{5} + \frac{u^3}{3} - u + \arctan(u) \right] + C\end{aligned}$$

Substitusikan kembali $u = t^{1/6}$:

$$= \frac{6}{7}t^{7/6} - \frac{6}{5}t^{5/6} + 2t^{1/2} - 6t^{1/6} + 6 \arctan(t^{1/6}) + C$$